

Evaluating generative antibody–drug conjugate linkers: chemical validity, covalent dynamics, and an FDA-anchored control

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ABSTRACT. Generative 3D models can now propose antibody–drug conjugate (ADC) linkers, shifting the open problem from *generation* to *evaluation*. We fine-tune DiffLinker on 73 ADC linker–payloads and show it learns ADC linker chemistry without memorization: cleavable motifs (Val-Cit, urea, Gly-Gly-Phe-Gly) emerge at 10–81 % versus 0–4 % in the zero-shot baseline, with median Tanimoto similarity 0.44–0.59 to the training set and zero near-copies across 1000 generations. The dominant practical failure is not the model but **chemical validity**: outputs routinely lose connectivity, valence, and stereochemistry, leaving only ≈ 10 fully 3D-valid candidates per 100 (Wilson 95 % CI 7.8–13.2) after a fail-closed validity gate. We then ask whether classical covalent molecular dynamics (MD) can *rank* the survivors. Building covalent trastuzumab-Fab/HER2 complexes at Cys214 and running 50 ns each, all candidates are **grossly stable** (covalent bond 1.81 Å, secondary structure preserved, per-domain C α RMSD 1.2–1.8 Å). An apparent payload-stability ranking is traced to a **topology artifact** — a pentavalent conjugation carbon — that vanishes and partially inverts once corrected. A pre-registered $N=3$ replicate study finds no separation (Welch $p=0.157$). Grafting the FDA-approved brentuximab-vedotin linker onto the identical scaffold yields a payload metric statistically indistinguishable from the generated candidates ($p=0.41, 0.20$). Fixed-bond MD payload RMSD is non-discriminative for ADC linkers because the chemistry that would discriminate them — succinimide hydrolysis, retro-Michael deconjugation, local pK_a — is absent from the model by construction.

1 Introduction

Antibody–drug conjugates (ADCs) couple the specificity of a monoclonal antibody to a cytotoxic payload through a chemical linker.[1] The linker is not a passive tether: its stability in circulation governs premature payload release, off-target toxicity, and ultimately the therapeutic index.[2] Designing a linker is therefore a multi-objective problem in which the hard constraint — plasma stability of the conjugation site — is precisely the property that is hardest to compute.

Recent work applies 3D generative diffusion models to *linker* design, holding the payload fixed and generating the spacer that connects it to the antibody-side conjugation handle.[3] These methods are improving quickly. What is rarely stress-tested is the *evaluation* step that follows generation — the procedure that is supposed to tell us which generated linker is best. In most pipelines it is assumed reliable: a potency surrogate, or a short molecular-dynamics (MD) simulation read out through a stability proxy such as payload root-mean-square deviation (RMSD).

This paper stress-tests that assumption, and it fails — cleanly, reproducibly, and against a clinical control. We report a complete generate $\rightarrow$ gate $\rightarrow$ build $\rightarrow$ simulate $\rightarrow$ evaluate pipeline (Figure 1) and show that while DiffLinker generates chemically plausible, novel ADC linkers, **neither a 2D potency classifier nor fixed-bond covalent MD can reliably rank them**. The strongest evidence is a positive control: replacing a generated linker with the authentic, FDA-approved brentuximab-vedotin linker and rerunning the identical simulation leaves the clinical linker statistically indistinguishable from our generated candidates.

Contributions.

- A reproducible DiffLinker $\rightarrow$ validity-gate $\rightarrow$ covalent-

MD pipeline for ADC linkers at a cysteine thiosuccinimide site (Figure 1).

- Evidence that **chemical validity — not generation — is the dominant practical bottleneck**, with a fail-closed valence/connectivity/stereo gate as the operative screen (Figure 2).
- A documented **ranking artifact**: a covalent-topology valence error produced a confident but spurious payload ranking that disappears under correction (Figure 4).
- A **pre-registered $N=3$ test plus an FDA positive control** showing fixed-bond MD payload RMSD does not discriminate generated linkers from one another or from an approved clinical linker (Figures 5–6).
- A cross-cutting lesson: **surface metrics (bond length, classifier AUC, score) mask structural errors; adversarial validation is required** (Table 3).

2 Background

3D diffusion for linker generation. DiffLinker[3] is an $E(n)$ -equivariant diffusion model pretrained on GEOM-Drugs[4] that generates a linker between fixed fragments. Its native output is atomistic — predicted 3D coordinates and atom types — without bond orders, perceived afterward from interatomic distances. This format is the origin of the chemical-validity problem of Section 4.2.

What sets ADC linker stability. The cysteine–maleimide adduct anchoring many clinical ADCs is metastable: the thiosuccinimide can undergo retro-Michael elimination and thiol exchange with plasma thiols (albumin, glutathione), releasing payload prematurely.[2] The competing, stabilizing pathway — hydrolytic ring-opening of the succinimide, catalysed by a proximal basic residue — renders the adduct resistant to retro-Michael.[5] Stability is thus

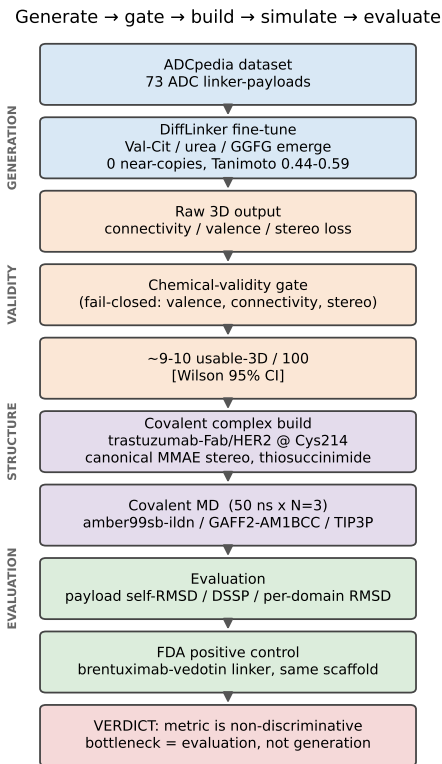


Figure 1. The generate → chemical-validity gate → covalent build → covalent MD → evaluation pipeline, with the FDA positive control.

controlled by bond-breaking chemistry, local pK_a , and the conjugation-site electrostatic microenvironment, none of which is represented in a classical fixed-bond force field.

Existing ADC scoring. Potency classifiers such as AMM predict cytotoxicity from a 2D conjugate representation plus biological context.[6] MD-derived geometric proxies are sometimes used as stability surrogates. Both assume the discriminating signal is captured; we test that directly.

3 Methods

3.1 Generation

DiffLinker was fine-tuned on 73 unique ADC linker-payloads spanning three conjugation chemistries (cysteine, lysine, click), recovered from the ADCpedia dataset[6] by a payload-anchored decomposition (coverage 98%). Two fine-tunes are reported (Table 1): a cysteine-only set (43 unique L/Ps) and the full multi-site superset (73 L/Ps, the figure quoted above). The candidates carried forward to covalent MD were generated from the best-yield cysteine-line checkpoint at the operational linker size; full dataset, sweep, and checkpoint details are in the SI. The numerically stabilized recipe (isfinite loss guard, restored pretraining optimizer state, learning rate 1×10^{-5} , batch 4, early stopping on denoising validation loss) is in the SI. Full generation methodology is supporting material — prior to this paper’s contribution.

3.2 Chemical-validity gate

A two-level gate. **Level A** (ligand, RDKit[7]): sanitizable; single connected component; no undefined stereocenters; correct maleimide/thiosuccinimide oxidation state. **Level B** (conjugate topology): per-atom bond count vs. element valence maximum, fail-closed at build time — the guard that auto-catches the over-valent conjugation carbon of Section 4.4.

3.3 Covalent complex construction

Receptor 1N8Z (trastuzumab Fab + HER2 ectodomain)[8], cleaned with PDBFixer (pH 7.4). The conjugation cysteine was chosen by side-chain SASA: **A:Cys214** (SG-SASA $43.75 \text{ \AA}^2$, the most exposed free cysteine). Because DiffLinker outputs lose stereochemistry through the coordinate → SDF step, canonical MMAE stereochemistry (10 stereocenters) was reimposed by CIP-driven iterative chiral-tag assignment (10/10 verified from 3D coordinates). Conjugation used a maleimide Michael addition; the ligand was aligned so the thioether carbon superimposed on the Cys214 SG ($C\text{-SG} = 1.80 \text{ \AA}$), with rotational clash minimization.

3.4 MD protocol

GROMACS 2025[9]; amber99sb-ildn[10] protein, GAFF2[11] with AM1-BCC charges[12] (acpype[13]) ligand, TIP3P water[14], 0.15 M NaCl, dodecahedral box (1.2 nm). The covalent C-SG linkage was a **true bond** (ligand merged into the light-chain moleculetype so 1-2/1-3 exclusions generate correctly), held by a harmonic term — a fixed-bond covalent model. Protocol: minimization → 100 ps NVT → 100 ps NPT (restrained) → 50 ns production at 2 fs. A fail-closed gate preceded every run: grompp clean, conjugation sulfur exactly two bonds ($C\beta$ + ligand carbon, no retained thiol proton), thioether sulfur type, conserved chain charge.

3.5 Evaluation metrics

Analyses used PBC-corrected trajectories (**-pbc whole→nojump**). We separate *integrity checks* (C-SG distance, ligand-protein minimum distance — bonded-topology-enforced, $\sigma \approx 0.04 \text{ \AA}$) from *behavioral observables*. Per-domain C α RMSD was self-fitted per domain — correct for a multi-domain Fab-HER2 assembly whose whole-complex RMSD is dominated by rigid-body motion. Secondary structure was tracked by DSSP.[15] Payload behavior used the **MMAE-core (51 heavy atoms) self-fitted RMSD**, isolating internal deformation from whole-ligand swing; the core was extracted by an order-verified substructure match that excludes the PABC carbamate carbon (which otherwise aliases the match).

3.6 Positive control

The authentic mc-Val-Cit-PAB-MMAE linker-payload of brentuximab vedotin (CAS 646502-53-6, $C_{68}H_{105}N_{11}O_{15}$, MW 1316.65) was conjugated to the **same** Cys214 on the **same** scaffold with the **same** thiosuccinimide chemistry, force field, and 50 ns × 3 protocol — everything constant except the linker.

Table 1. ADC-motif emergence and novelty, fine-tuned vs. zero-shot (matched size). Zero near-copies (Tanimoto > 0.85) per 1000 generations across all fine-tuned configurations.

Model	size	Val-Cit	Urea	med. Tan.
Zero-shot	20	0 %	0 %	0.30
Fine-tuned (cys)	20	23 %	35 %	0.58
Fine-tuned (cys)	60	64 %	81 %	0.44
Fine-tuned (multi)	20	10 %	13 %	0.59
Fine-tuned (multi)	60	43 %	69 %	0.46

3.7 Statistical design (pre-registered)

$N=3$ independent replicates per system (rep 1 continued from NPT velocities; reps 2–3 fresh velocity draws, distinct seeds, `continuation=no`). Metric: MMAE-core self-RMSD over 25–50 ns. Pre-registered separation criterion: mean $\pm$ SD ranges disjoint **and** Welch $p < 0.05$.

4 Results

4.1 DiffLinker learns ADC chemistry without memorizing

After fine-tuning, the model produces cleavable-linker signatures absent from the pretrained baseline: Val-Cit ureido, urea, and Gly-Gly-Phe-Gly motifs emerge at 10–81 % versus 0–4 % zero-shot at matched size (Table 1). This is learning, not copying: across 1000 generations there were **zero near-copies** (Tanimoto > 0.85), and median Tanimoto to the training set was 0.44–0.59. The fine-tune also unlocked generation at ADC linker length, which the zero-shot model could not reach. (Two intrinsic tradeoffs — chemistry vs. connectivity along size, and target-specificity vs. topological diversity along the training distribution — are characterized in the SI; they do not affect the conclusions here.)

4.2 Chemical validity, not generation, is the bottleneck

DiffLinker’s atomistic output systematically fails three chemical-validity criteria. *Connectivity*: only ≈ 30 –45 % of outputs are a single connected fragment at the operational size. *Valence*: bond perception leaves under- or over-coordinated atoms. *Stereochemistry*: the coordinate $\rightarrow$ SDF conversion strips stereochemistry irrecoverably, so the active configuration of the payload (e.g. MMAE’s 10 stereocenters) must be reimposed explicitly. After the fail-closed gate, only ≈ 10 **fully 3D-valid candidates survive per 100 generated** (Wilson 95 % CI 7.8–13.2; Figure 2). An industrial campaign must therefore oversample by an order of magnitude, and the validity gate — not the generator — is the operative screen: a millisecond-cost gate does work the MD cannot.

4.3 Gross MD stability: all candidates pass

All four lead candidates are stable over 50 ns at the coarse level (Figure 3, Table 2). The covalent thioether holds at a mean C–SG of 1.81 Å (no drift) in every run. Per-domain C α RMSD shows the Fab is internally rigid (light/heavy chain 1.2–1.6 Å), with elevated values confined to the intrinsically flexible HER2 peripheral domains I and IV

Chemical validity — not generation — is the bottleneck

usable-3D = 10.2 / 100 [95% CI 7.8–13.2]

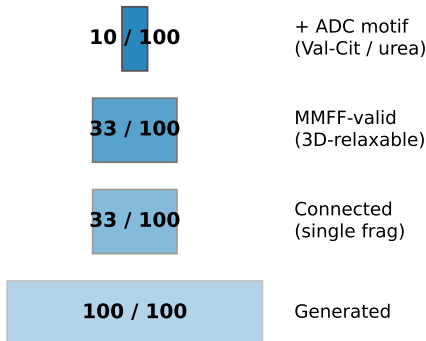


Figure 2. Usable-3D yield funnel. After the fail-closed gate, ≈ 10 of every 100 generated molecules are connected, valence-correct, stereo-correct, and ADC-motif-bearing.

Table 2. Gross MD stability, per candidate (means over 25–50 ns, computed from the trajectories). LC/HC = light/heavy chain; HER2 = whole ectodomain. All verdicts: **stable**.

Cand. (output)	C–SG (Å)	DSSP (%)	LC/HC (Å)	HER2 (Å)	verdict
4 (486)	1.81	43.4 $\pm$ 0.9	1.2/1.3	3.8	stable
1 (219)	1.81	43.4 $\pm$ 1.1	1.5/1.6	3.8	stable
5 (345)	1.81	43.4 $\pm$ 0.9	1.3/1.6	2.9	stable
8 (106)	1.81	43.5 $\pm$ 0.9	1.9/1.6	3.8	stable

(Figure 3a). Secondary structure is preserved: DSSP-structured fraction stays at ≈ 43 % (SD < 1.1 %; Figure 3b), the decisive test against unfolding. The payload stays attached and in contact; no dissociation occurs. Candidates 1 and 8 (Table 2) retain the first-pass junction topology later corrected in Section 4.4; this does not affect their gross-stability classification, because the C–SG distance (a harmonic term), the per-domain C α RMSD, and DSSP are all determined away from the ligand junction and are invariant to its valence — the defect perturbs only the payload’s internal self-RMSD. The discrimination analysis (Section 4.4 onward) is therefore restricted to the valence-corrected candidates 4 and 5 plus the control. At the coarse level the model behaves correctly — which sets up the discrimination question: can it tell the candidates apart?

4.4 The ranking artifact

It initially appeared so. On the first-pass topologies, candidate 4 looked markedly more rigid than candidate 5 (MMAE-core self-RMSD 1.21 vs. 4.26 Å; Figure 4, grey), with a more compact payload and fewer contacts — a clean apparent ranking. Adversarial inspection of the junction revealed the cause: the thioether bond had been added to the ligand carbon **without removing one hydrogen**, producing a **pentavalent carbon**. GROMACS does not check valence, so it ran; a distance-only check passed because the harmonic bond holds its length regard-

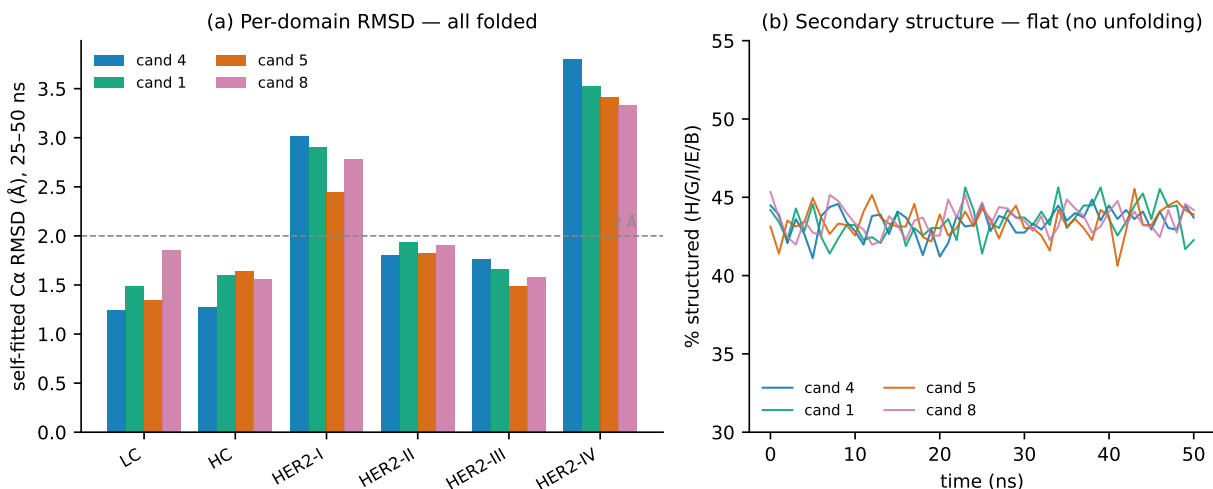


Figure 3. Gross MD stability (50 ns). (a) Per-domain self-fitted Cα RMSD: the Fab is internally rigid; elevated RMSD is confined to HER2 peripheral domains I and IV. (b) DSSP structured fraction is flat ($\approx 43\%$, $SD < 1.1\%$) for all four candidates — no unfolding.

less. The over-constrained junction clamped the payload, manufacturing the apparent rigidity.

Correcting the valence (deleting one hydrogen, restoring a tetrahedral CH(S) thioether) erases the ranking. Candidate 4’s self-RMSD rises from 1.21 to 3.51 Å — into candidate 5’s range — and the secondary metrics invert: candidate 5, which had looked worst, becomes *more* compact (R_g 6.36 $\rightarrow$ 4.77 Å) and *less* adsorbed (contacts 504 $\rightarrow$ 96) than candidate 4 (Figure 4, colour). The apparent “candidate 4 > candidate 5” ranking was a chemistry-bug artifact, not a molecular property. (This was the second instance of the same failure mode: an earlier defect left the cysteine thiol proton on the conjugated sulfur, also passing a distance-only check — Table 3.)

4.5 Replicates: no separation

With correct chemistry, do the candidates separate beyond run-to-run noise? Three independent replicates (Figure 5) give candidate 4 = 3.15 ± 0.96 Å and candidate 5 = 4.33 ± 0.30 Å. The means differ by 1.18 Å, but Welch gives $t = -2.04$, $df = 2.40$, $p = 0.157$, with overlapping ranges — **no separation**. Candidate 4’s intra-candidate spread (SD 0.96 Å, replicates 2.06–3.87 Å) is as large as the inter-candidate gap: the payload is not configurationally converged at 50 ns, so a self-RMSD-vs-start “ranking” reads drift from an arbitrary pose. $N=3$ is low-powered ($df \approx 2.4$) and detects only a large effect — it does not prove the candidates identical; it proves no signal survives noise at this cost.

4.6 Positive control: an FDA linker is indistinguishable

The decisive test removes the power objection. Grafting the FDA-approved brentuximab-vedotin linker onto the identical scaffold and rerunning the identical $N=3$ protocol gives a payload self-RMSD of 3.75 ± 0.54 Å — landing **between** the two candidates, inside their envelope (Figure 6), non-significant against both (vedotin vs. candidate 4 $p = 0.41$; vs. candidate 5 $p = 0.20$).

Key result. A metric that cannot distinguish a clinically approved ADC linker from arbitrary generated candidates is non-discriminative *by demonstration* — independent of statistical power. Encouragingly, the generated candidates occupy the same payload-dynamics regime as an approved ADC, so they are physically realistic.

4.7 Why the metric fails — by construction

The failure is structural, not a sampling deficiency to be cured with more nanoseconds. The model has a fixed harmonic covalent bond, no bond-breaking, no constant-pH, and no plasma thiols — it omits exactly the chemistry that sets ADC stability: succinimide ring-opening, retro-Michael, local pK_a . Consistent with this, static + dynamic analysis of the Cys214 site (Figure 7) finds it acidic near the sulfur ($pK_a \approx 9.9$; nearest base Arg211 at 8.9 Å, beyond catalytic range), with only transient, flexible-linker-driven basic contacts — weak self-stabilization. But the site is **shared across all candidates**, so it annotates the site, not the linker. *The quantity that would rank candidates is precisely the quantity the cheap model omits.*

The 2D potency classifier (AMM) fails to rank for a parallel reason: its label is saturated. A potent payload (MMAE) on an amplified antigen (HER2) at a 10 nM cytotoxicity threshold is predicted active for essentially any reasonable linker, and the model — once a silent checkpoint-loading defect was corrected (Table 3) — assigns near-saturated scores even to junk controls. Dimensionality is not the bottleneck; the label is.

5 Discussion

The bottleneck moved from generation to evaluation.

Generation is solved enough for practical use, the residual failure (chemical validity) handled by a cheap gate. What is *not* solved is ranking: two independent evaluation routes

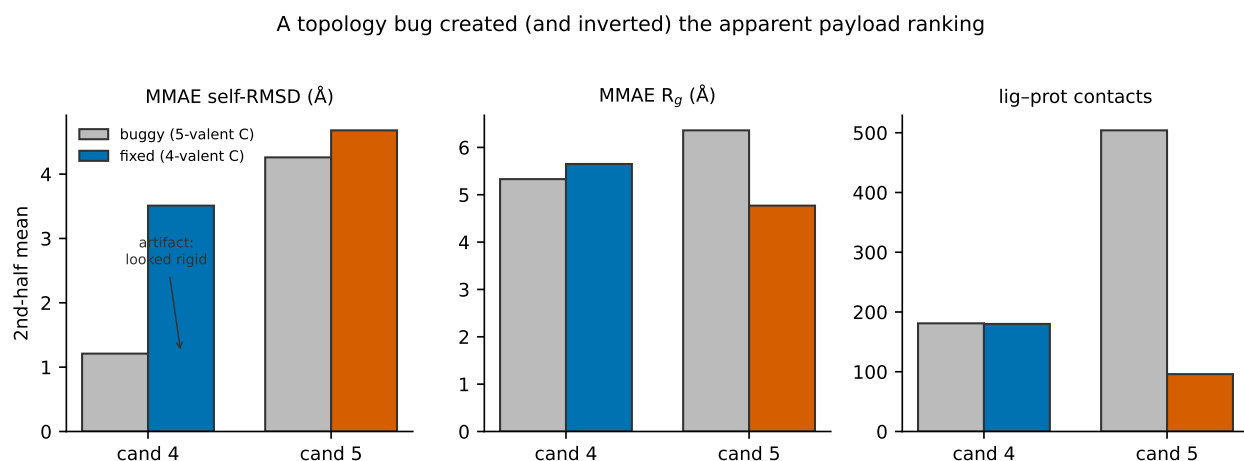


Figure 4. The ranking artifact. Buggy topology (pentavalent conjugation carbon, grey) vs. corrected tetravalent thioether (colour). The apparent “candidate 4 more rigid” ranking disappears (self-RMSD) and the secondary metrics (R_g , contacts) invert once the valence is corrected.

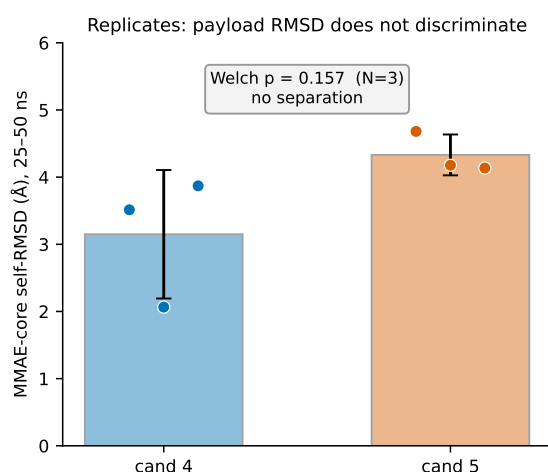


Figure 5. $N=3$ replicates, candidates 4 vs. 5. Points are replicates, bars mean $\pm$ SD. No separation (Welch $p = 0.157$); intra-candidate spread $\approx$ inter-candidate gap.

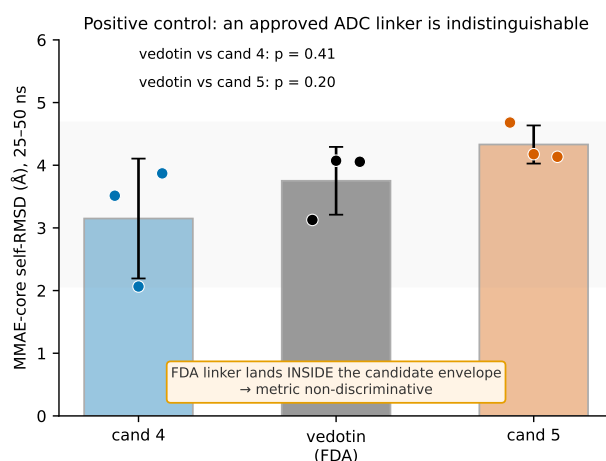


Figure 6. Positive control. The FDA-approved brentuximab-vedotin linker (black), conjugated to the same scaffold and run identically, gives a payload self-RMSD inside the generated-candidate envelope and statistically indistinguishable from both ($p = 0.41, 0.20$).

— a 2D potency classifier and fixed-bond covalent MD — both fail to discriminate, and the MD route fails against a clinical control.

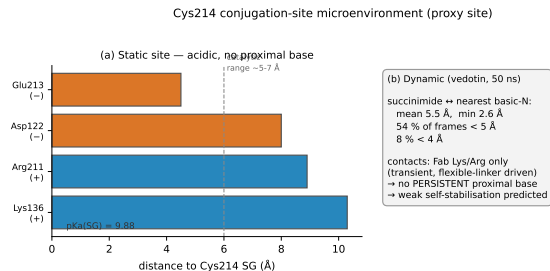
Surface metrics mask structural errors. The recurring lesson is that an easily-read quantity can pass while the object it summarizes is wrong (Table 3): a classifier loaded non-strict reported high AUC on partly random weights; a covalent topology held its bond length while carrying an impossible valence; a payload “looked rigid” because a bug over-constrained it; small-scale fine-tuning showed yield differences within run-to-run variance. Each was caught only by adversarial validation — negative controls, valence assertions, multi-seed evaluation with confidence intervals. Validating the object rather than the proxy should be the default.

Ranking sites and ranking linkers are different problems. Ranking *conjugation sites* (Cys214 vs. a hinge Cys vs. a lysine) needs site diversity and is approachable with structure-derived descriptors. Ranking *linkers at a fixed site* needs the deconjugation chemistry — reactive or constant-pH MD, or experimental labels. Because every candidate sits on the same Cys214, no site descriptor can separate them, and fixed-bond dynamics cannot either.

What a valid evaluation would require. Two routes. (i) *Data*: experimental deconjugation/plasma-stability measurements with known site and linker; the public literature is small, fragmented across system and assay matrix, and not poolable — a hard data wall. (ii) *Physics*: constant-pH MD (local pK_a) is the most tractable next lever and, unlike a static descriptor, varies per linker

Table 3. Methodological findings — a surface metric that passed vs. the structural error it masked.

#	Surface metric that passed	Structural error it masked	Caught by
1	Classifier validation AUC	Broken checkpoint (silent <code>strict=False</code>)	Negative-control SMILES
2	Renamed-keys load “succeeds”	Vestigial untrained head loaded as operative one	BatchNorm tracking counters
3	Conjugate builds and runs	Stereochemistry silently stripped	CIP check vs. canonical payload
4	Covalent bond length 1.81 Å	Retained thiol proton → hypervalent S	Valence assertion
5	Covalent bond length 1.81 Å	Pentavalent conjugation carbon	Valence assertion
6	Trajectory “looks centered”	Multi-chain PBC error → multi-nm RMSD	<code>-pbc whole→nojump</code>
7	Single-checkpoint yield gain	GPU run-to-run trajectory variance	Multi-seed + Wilson CI

**Figure 7.** Cys214 microenvironment. (a) Static: the immediate SG environment is acidic; the nearest basic residue is beyond catalytic range. (b) Dynamic (vedotin, 50 ns): only transient basic contacts. Weak self-stabilization — and, being shared across candidates, the site cannot rank linkers.

through each maleimide’s local solvation; reactive MD of retro-Michael is research-grade. What is *not* justified is more sampling of the present fixed-bond model — longer trajectories add wall-clock to an incomplete physics, not the reaction.

6 Limitations

For a negative result these are the substance, not a coda.

- **Statistical power.** $N=3$ detects only large effects; the result is “no separation survives noise at this cost,” not “molecules are identical.” The positive control, not the p -value, carries the claim.
- **Proxy site.** 1N8Z lacks the IgG hinge cysteine of clinical brentuximab vedotin; Cys214 characterizes the *modeling* site.
- **FF/chemistry approximations.** GAFF2/AM1-BCC payload; single arbitrary thiosuccinimide diastereomer; basic amines (candidates 1, 4) modeled neutral; 300 K; bond-only junction.
- **Configurational equilibration.** Thermodynamically equilibrated, but the slow payload-surface mode is not converged at 50 ns — itself part of why self-RMSD is fragile.

7 Crucial next steps

This work establishes that the bottleneck is evaluation, not generation, and that the discriminating quantity is absent from both rankers we tested. Countering each proven

limitation requires, in priority order:

1. **Put the deconjugation chemistry into the simulation.** (Counters the fixed-bond MD failure, §4.6–4.7.) Replace the harmonic covalent bond with (i) constant-pH MD to capture the per-linker local pK_a /protonation that governs succinimide ring-opening — tractable, the single best next lever — and (ii) QM or reactive-MD estimates of the retro-Michael and hydrolysis barriers (research-grade). Pilot on candidate 4 and the vedotin control: does the barrier or local pK_a separate what self-RMSD could not?
2. **Acquire experimental deconjugation labels.** (Counters the data wall, §5.) No stability predictor is falsifiable without them. Tier 1: curate the few literature ADCs with known antibody/site/linker plus a plasma-stability readout (qualitative trends only — matrices do not pool). Tier 2: a controlled wet-lab series (one antibody, one payload, systematically varied linker, measured % payload retained / $t_{1/2}$) — the real unlock, requiring an experimental collaboration. Data, not method, is the gating reagent.
3. **Fix the potency oracle’s label, not its dimensionality.** (Counters AMM saturation, §4.7.) The 10 nM cytotoxicity label is linker-insensitive for potent payloads on amplified antigens. Re-head or retrain AMM on a linker-discriminating endpoint (lower threshold, plasma stability, or pharmacokinetics); otherwise use it strictly as a plausibility filter, never a ranker.
4. **Rank sites, where descriptors actually work.** (Counters the site/linker conflation, §5.) A microenvironment descriptor cannot rank linkers on a shared site, but it can rank *conjugation sites* given site diversity. Build a numerical site-stability descriptor (SG SASA, local pK_a , proximal-base distance, electrostatic microenvironment) across Cys214, the IgG hinge cysteine, and lysine sites, anchored to clinically known stable and unstable sites.
5. **Release the validation discipline as a tool.** (Counters the surface-metric trap, Table 3.) The chemical-validity gate, the covalent-topology valence assertions, and multi-seed + Wilson-CI evaluation caught every artifact in this study; packaged as a reusable pre-screen,

they would make generated molecules validated by construction rather than by luck.

What we will *not* do: more or longer fixed-bond MD of the same system — it adds wall-clock to an incomplete physics, not the missing reaction.

8 Conclusion

DiffLinker generates plausible, novel ADC linkers; chemical validity, handled by a cheap fail-closed gate, is the practical bottleneck of generation. But neither a 2D potency classifier (label-saturated) nor fixed-bond covalent MD (chemistry-omitting) can reliably rank them — demonstrated against an FDA-approved linker the MD metric cannot distinguish from arbitrary candidates. The next advance will not come from a better generator; it will come from an evaluation model grounded in conjugation-site chemistry — succinimide hydrolysis, retro-Michael kinetics, local pK_a — and/or in experimental deconjugation data.

Acknowledgements

Funding and compute resources: TBD.

Data and code availability

Validity gate, conjugation/stereo and MD scripts, figure code (`paper/figures/make_figures.py`), and complex-only trajectories: repository TBD. Receptor: PDB 1N8Z. Control linker: brentuximab vedotin (CAS 646502-53-6).

References

- [1] Joshua Z. Drago, Shanu Modi, and Sarat Chandarlapaty. Unlocking the potential of antibody–drug conjugates for cancer therapy. *Nature Reviews Clinical Oncology*, 18: 327–344, 2021. doi: 10.1038/s41571-021-00470-8.
- [2] Ben-Quan Shen et al. Conjugation site modulates the in vivo stability and therapeutic activity of antibody–drug conjugates. *Nature Biotechnology*, 30:184–189, 2012. doi: 10.1038/nbt.2108.
- [3] Ilia Igashov, Hannes Stärk, Clément Vignac, Arne Schneuing, Victor Garcia Satorras, Pascal Frossard, Max Welling, Michael Bronstein, and Bruno Correia. Equivariant 3D-conditional diffusion model for molecular linker design. *Nature Machine Intelligence*, 6:417–427, 2024. doi: 10.1038/s42256-024-00815-9.
- [4] Simon Axelrod and Rafael Gómez-Bombarelli. GEOM, energy-annotated molecular conformations for property prediction and molecular generation. *Scientific Data*, 9: 185, 2022. doi: 10.1038/s41597-022-01288-4.
- [5] Robert P. Lyon et al. Self-hydrolyzing maleimides improve the stability and pharmacological properties of antibody–drug conjugates. *Nature Biotechnology*, 32:1059–1062, 2014. doi: 10.1038/nbt.2968.
- [6] Hazem Mslati et al. ADCpedia: a multimodal model for antibody–drug conjugate potency prediction. *bioRxiv*, 2026. doi: 10.64898/2026.05.22.727320. Preprint.
- [7] RDKit. RDKit: Open-source cheminformatics. <https://www.rdkit.org>. Accessed 2026.
- [8] Hyun-Soo Cho, Karen Mason, Kasra X. Ramyar, Ann Marie Stanley, Sandra B. Gabelli, Dan W. Denney, and Daniel J. Leahy. Structure of the extracellular region of HER2 alone and in complex with the Herceptin Fab. *Nature*, 421:756–760, 2003. doi: 10.1038/nature01392.
- [9] Mark James Abraham, Teemu Murtola, Roland Schulz, Szilárd Páll, Jeremy C. Smith, Berk Hess, and Erik Lindahl. GROMACS: High performance molecular simulations through multi-level parallelism from laptops to supercomputers. *SoftwareX*, 1–2:19–25, 2015. doi: 10.1016/j.softx.2015.06.001.
- [10] Kresten Lindorff-Larsen, Stefano Piana, Kim Palmo, Paul Maragakis, John L. Klepeis, Ron O. Dror, and David E. Shaw. Improved side-chain torsion potentials for the Amber ff99sb protein force field. *Proteins: Structure, Function, and Bioinformatics*, 78:1950–1958, 2010. doi: 10.1002/prot.22711.
- [11] Junmei Wang, Romain M. Wolf, James W. Caldwell, Peter A. Kollman, and David A. Case. Development and testing of a general Amber force field. *Journal of Computational Chemistry*, 25:1157–1174, 2004. doi: 10.1002/jcc.20035.
- [12] Araz Jakalian, David B. Jack, and Christopher I. Bayly. Fast, efficient generation of high-quality atomic charges. AM1-BCC model: II. parameterization and validation. *Journal of Computational Chemistry*, 23:1623–1641, 2002. doi: 10.1002/jcc.10128.
- [13] Alan W. Sousa da Silva and Wim F. Vranken. ACPYPE – AnteChamber PYthon Parser interface. *BMC Research Notes*, 5:367, 2012. doi: 10.1186/1756-0500-5-367.
- [14] William L. Jorgensen, Jayaraman Chandrasekhar, Jeffrey D. Madura, Roger W. Impey, and Michael L. Klein. Comparison of simple potential functions for simulating liquid water. *The Journal of Chemical Physics*, 79:926–935, 1983. doi: 10.1063/1.445869.
- [15] Wolfgang Kabsch and Christian Sander. Dictionary of protein secondary structure: pattern recognition of hydrogen-bonded and geometrical features. *Biopolymers*, 22:2577–2637, 1983. doi: 10.1002/bip.360221211.